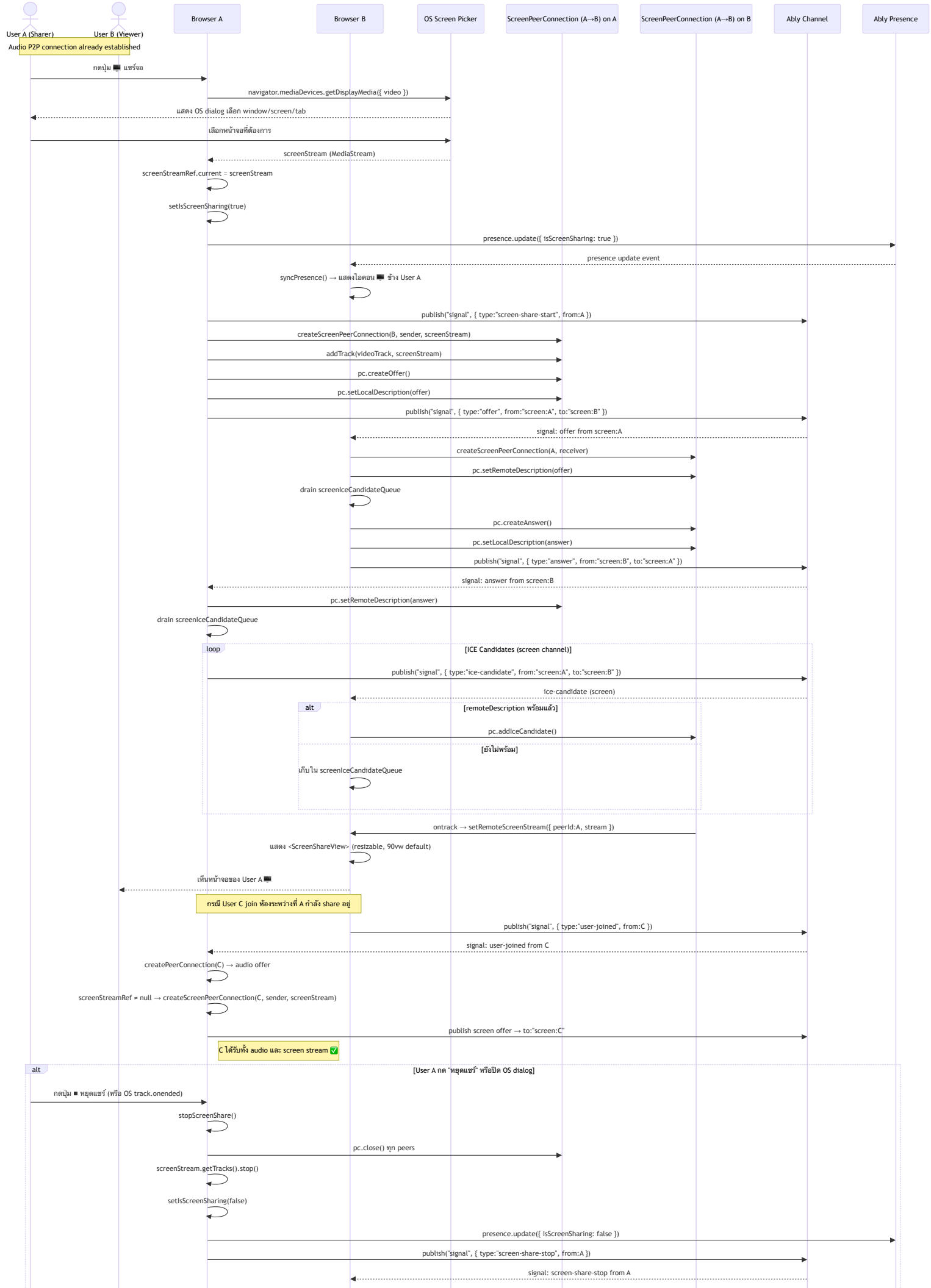
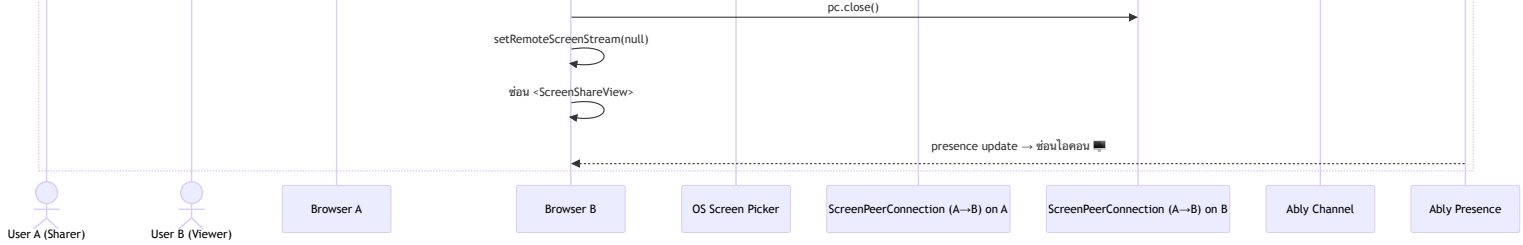

User A (Sharer)
User B (Viewer)
Browser A
Browser B
OS Screen Picker
ScreenPeerConnection (A→B) on A
ScreenPeerConnection (A→B) on B
Ably Channel
Ably Presence
Audio P2P connection already established
กดปุ่ม 🖥️ แชร์จอ
navigator.mediaDevices.getDisplayMedia({ video })
แสดง OS dialog เลือก window/screen/tab
เลือกหน้าจอที่ต้องการ
screenStream (MediaStream)
screenStreamRef.current = screenStream
setIsScreenSharing(true)
presence.update({ isScreenSharing: true })
presence update event
syncPresence() → แสดงไอคอน 🖥️ ข้าง User A
publish("signal", { type:"screen-share-start", from:A })
createScreenPeerConnection(B, sender, screenStream)
addTrack(videoTrack, screenStream)
pc.createOffer()
pc.setLocalDescription(offer)
publish("signal", { type:"offer", from:"screen:A", to:"screen:B" })
signal: offer from screen:A
createScreenPeerConnection(A, receiver)
pc.setRemoteDescription(offer)
drain screenIceCandidateQueue
pc.createAnswer()
pc.setLocalDescription(answer)
publish("signal", { type:"answer", from:"screen:B", to:"screen:A" })
signal: answer from screen:B
pc.setRemoteDescription(answer)
drain screenIceCandidateQueue
loop
[ICE Candidates (screen channel)]
publish("signal", { type:"ice-candidate", from:"screen:A", to:"screen:B" })
ice-candidate (screen)
alt
[remoteDescription พร้อมแล้ว]
pc.addIceCandidate()
[ยังไม่พร้อม]
เก็บใน screenIceCandidateQueue
ontrack → setRemoteScreenStream({ peerId:A, stream })
แสดง <ScreenShareView> (resizable, 90vw default)
เห็นหน้าจอของ User A 🖥️
กรณี User C join ห้องระหว่างที่ A กำลัง share อยู่
publish("signal", { type:"user-joined", from:C })
signal: user-joined from C
createPeerConnection(C) → audio offer
screenStreamRef ≠ null → createScreenPeerConnection(C, sender, screenStream)
publish screen offer → to:"screen:C"
C ได้รับทั้ง audio และ screen stream ✅
alt
[User A กด "หยุดแชร์" หรือปิด OS dialog]
กดปุ่ม ■ หยุดแชร์ (หรือ OS track.onended)
stopScreenShare()
pc.close() ทุก peers
screenStream.getTracks().stop()
setIsScreenSharing(false)
presence.update({ isScreenSharing: false })
publish("signal", { type:"screen-share-stop", from:A })
signal: screen-share-stop from A


pc.close()
setRemoteScreenStream(null)
ซ่อน <ScreenShareView>
presence update → ซ่อนไอคอน 🖥️
User A (Sharer)
User B (Viewer)
Browser A
Browser B
OS Screen Picker
ScreenPeerConnection (A→B) on A
ScreenPeerConnection (A→B) on B
Ably Channel
Ably Presence